Macro , Macro Placements & its Guidelines

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What Are Macros ?

- Specially designed large cells in ASIC design.
- Typically **memories, analog blocks, IP cores.**
- Proper placement of macros has **direct impact** on chip quality & performance.

Types of Macros

1. Hard Macros

1. Circuit is **fixed** (layout is predefined)
2. Functionality is not visible
3. Only **timing & physical info** is available

2. Soft Macros

1. Circuit is **not fixed** (flexible netlist)
2. Functionality & gate-level info is visible
3. Timing info is available and can be optimized

VLSIGURU

Basic constraints

- Alignment of macros by edges.
- Group macros logically, leaving large **rectangular areas** for standard cell placement
- Alignment of macros with core boundary.

Goals of Macros Placement

Better Rout ability

Timing

Reduced Wire length

Area for standard cells

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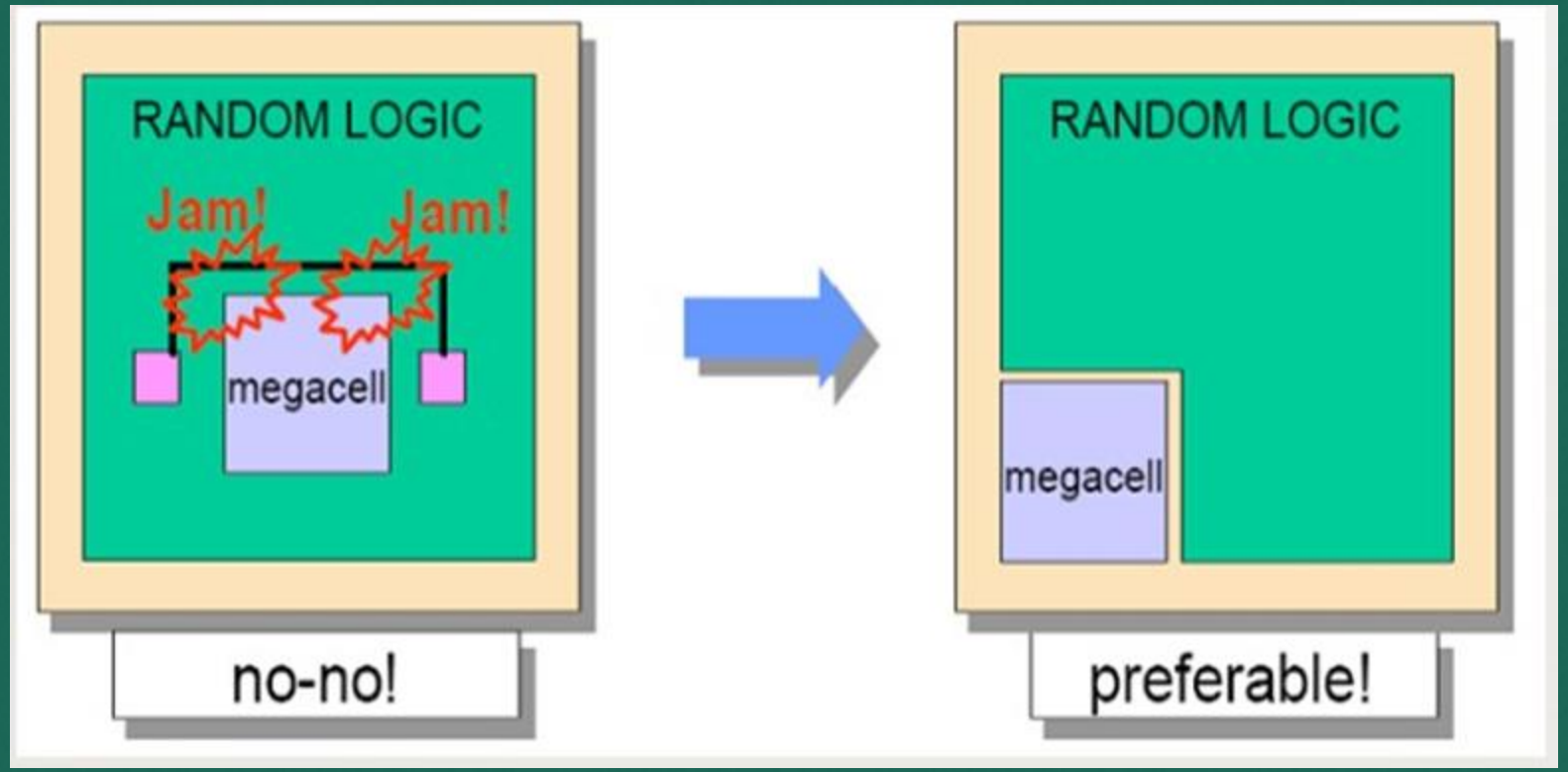
Macro Placement Guidelines

1. Macros should be placed at the periphery of the block. Avoid placement of Macros or stacks near the ports in order to avoid congestion in later stages of physical design.

2. Avoid stacking of Macros particularly near to ports.

   >> Stacking of macros results in CONGESTION.

Congestion = routing traffic jam → harder to close timing, larger chip area, and more power consumption.


RANDOM LOGIC
Jam!
Jam!
megacell
no-no!
RANDOM LOGIC
megacell
preferable!

3. Placement of macros should happen in a way so that

Macro pins face towards core to avoid detour.

4. Place Macro's based on Flyline analysis

FlyLine : It shows logical connection between :

IO ports ↔ macros

Macro ↔ macro

Macro ↔ standard cells

Types : Data Flow FlyLines, Net FlyLines

macros to macros fly lines:

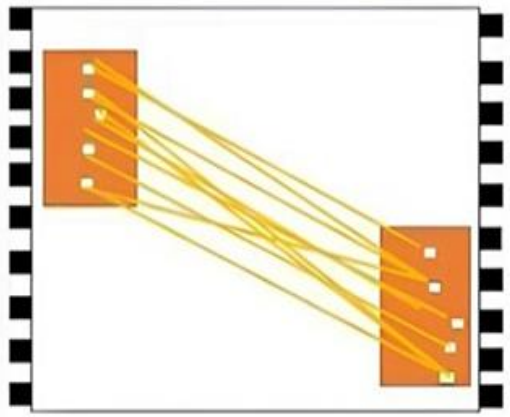


Fig a: macro to macro fly lines

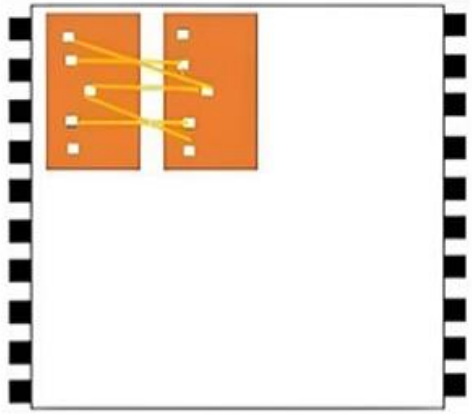


fig b: macros placed near to each other

macros to IO pin:

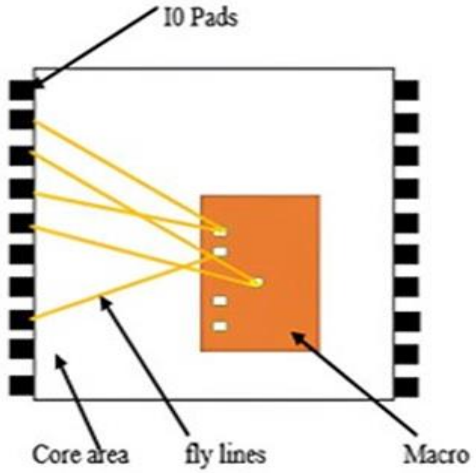


Fig a: macro to IO ports fly lines

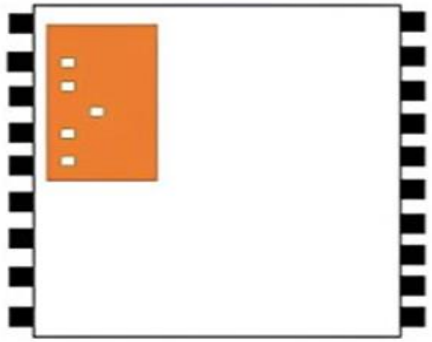
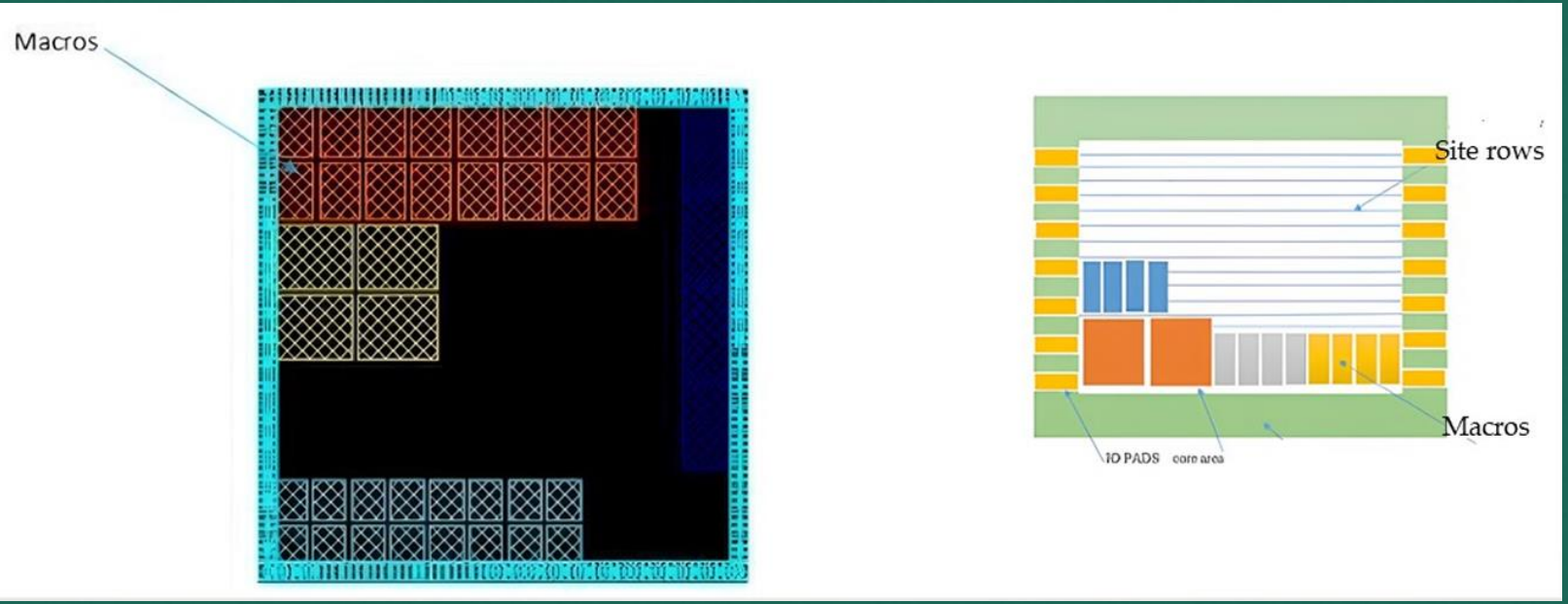


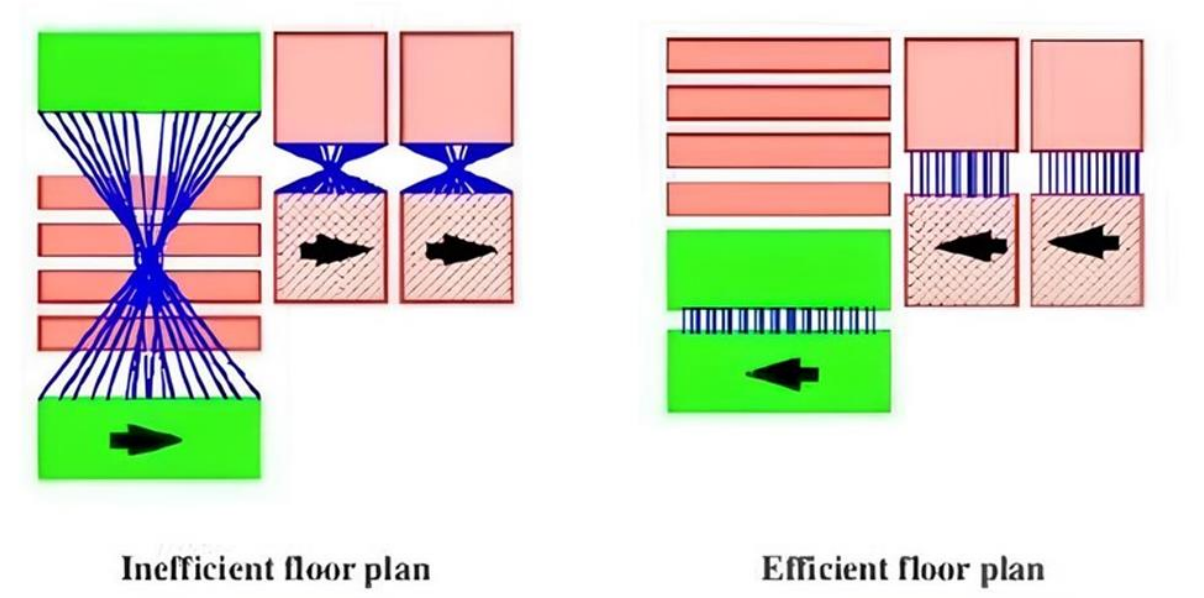
Fig b: macro placed at the core boundary

5. Place macros based on families or Hierarchy:

i.e., macros belonging to same family place together.

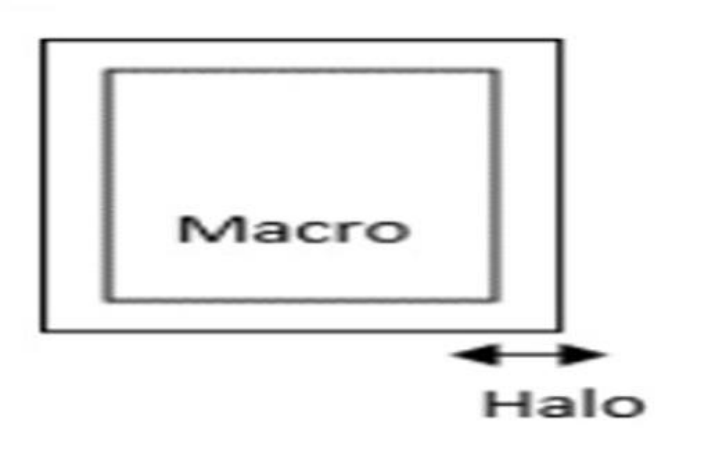


6. Macros should be placed by taking care of channel optimization to avoid CrissCross placement of Macros.



7. Placement should be done in proper multiple of tracks so that macros can be routed properly.

8. After macro placement, we need to place halos/keepout margins around the Macros to avoid congestion around the macros.



9. Placement blockages are applied between macro channel, so that no standard cell should sit in the channel, in order to avoid routing congestion.

10. There should be minimum gap between the Macros.

$$\text{Distance between Macros (Spacing)} = \frac{(\text{No. of pins facing channel}) \times (\text{Pitch of metal})}{\text{No. of routing layers}}$$

We should allow channels for routing pins and for buffer insertion which helps during timing optimization while placing possible required buffer.

Thank You

